Sanandan Ojha

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RESEARCH SUMMARY

Undergraduate researcher interested in exploring interdisciplinary problems in biomedical sciences using engineering and mathematics. Experienced with continuum mechanics, fluid dynamics, medical imaging, and parameter estimation. Looking to develop skills in numerical methods theory and design, especially for non-linear partial differential equations. Eager to contribute computational and engineering skills towards translational biomedical research.

EDUCATION

University of Florida, Gainesville FL: 3.91 GPA *August 2023—May 2027*

Bachelor of Science, Mathematics
Bachelor of Science, Biomedical Engineering
Minor, Statistics

RESEARCH EXPERIENCES

University of Florida, Honors Thesis
Advisor: Dr. Stepien, Dr. Murfee *August 2026—Present*

- Using stress profiles of microcirculatory slides to estimate clearance in the livers. Work in progress.

University of Florida, Mathematical Biology
Advisor: Dr. Stepien *October 2025—Present*

- Developing a **2**-layer continuum mechanics model for cell migration in gastrulation stage of development.
- Using **UQ-Lab** and MATLAB to conduct Sensitivity Analysis and Approximate Bayesian Computation – Sequential Monte Carlo (**ABC-SMC**) to fit parameters and models to experimental data on HiPerGator.

University of Florida, Systems Medicine
Advisor: Dr. Brunson *May 2026—Present*

- Developing a novel mechanistic model for primary graft dysfunction in lung transplantations.
- Utilizing pathway mining and conceptual modeling to develop the model.

Florida Theragnostic Center
Advisor: Dr. Krishnan *May 2026—May 2026*

- Case study on Xofigo (Dichloride) imaging and therapy in patients.
- Demonstrated a correlation between Bone Scan Index and patient outcomes.

University of Florida, International Genetic Engineering Machine (iGEM)
Advisor: Dr. Pan *February 2025—October 2025*

- Utilized AI to design a novel binder targeting osteocalcin, including **BindCraft** workflow on HiPerGator.
- Contributed to project documentation using **React** to receive **Silver** recognition at 2025 Grand Jamboree.

Advisor: Dr. Pan *February 2024—October 2024*

- Adapted a differential model of sepsis in bone marrow into an agent-based model using **NetLogo.**
- Contributed to project documentation using HTML to receive **Gold** recognition at 2024 Grand Jamboree.

Indiana University Indianapolis, Research Experience for Undergraduates
Advisor: Dr. Zhu *May 2025—July 2025*

- Developed a Lattice-Boltzmann Immersed-Boundary (**LB-IB**) method for cells in low Reynolds number flows.
- Increased efficiency by **300 steps/sec** with minimal error **1E-9 per timestep** over traditional LB-IB.
- Constructed an implicit LB-IB framework for cell migration in low Reynolds number scenarios.

University of Florida, Dream Team Engineering

Advisor: Dr. Shah *January 2025—Present*

- Testing Liver, Kidney, Cholecystectomy, and Caudal Block **surgical simulators** using medical residents/students.
- Gathering data to establish the efficacy of simple surgical models in giving residents hands-on experience.

Advisor: Dr. Wong *August 2024—May 2025*

- Using **Diffusion ML models** to generate MRI images from CTs for perioperative planning of patients undergoing Deep Brain Stimulation and contraindicated for MRIs.

Advisor: Dr. Chandran *September 2023—January 2025*

- Constructed **experimental design** for efficacy of a 3D printed Berlin Heart and supraventricular tachycardia patient education models.

University of Florida, Biochemistry

Advisor: Dr. Caglayan *January 2024—May 2025*

- **Overexpression** and **purification** of recombinant human DNA repair proteins.
- Biochemical and structural characterization of DNA repair proteins.

University of Pittsburgh, Radiology

Advisor: Dr. Muthukrishnan *May 2022—June 2022*

- Collated clinical and imaging data to evaluate Prostate Specific Membrane Antigen (PSMA) imaging efficacy.
- Established a correlation between patient outcomes and PSMA PET performance relative to conventional PET.

WORKSHOPS AND ENRICHMENT

University of Arizona, Preliminary Arizona Winter School (PAWS) *September 14, 2026—Present*

Coordinator: Dr. Brandon Levin

- Attending lectures on Quaternion Algebra by Dr. Jen Berg.
- Weekly problem set completed in collaboration with sessions led by Dr. Divyasree C-Ramachandran.

University of Florida, Directed Reading Program *August 31, 2026—Present*

Coordinator: Dr. Sara Pollock

- Reading “Weak Convergence Methods for Non-linear Partial Differential Equations” by Evans.
- Attending weekly guided session by Connell Dylan, PhD candidate.

PI3, Polyploidy Undergraduate Research Experience *June 29—July 1, 2026*

Coordinator: Dr. Pam Soltis

- Attended lectures on transcription analysis, proteomics analysis, and polyploidy investigation.

Texas A&M, Modeling and Simulating with Partial Differential Equations (PDEs) *May 18—May 29, 2026*

Coordinator: Dr. Matthias Maier

- Attended lectures on PDE theory, Finite Difference and Finite Element methods, and Transport Equations.
- Created a solver for the Radiation Transport Equation directed by Dr. Germond.

University of Florida, Infectious Diseases and Economic Growth *March 26—March 29, 2026*

Coordinator: Dr. Calistus Ngonghala

- Attended lectures on modeling infectious disease spread and interactions with the economy.
- Designed a model for early COVID-19 spread in Florida, including parameter estimation using ABC.

POSTER PRESENTATIONS

Ojha, Sanandan; *Dadlani, Rishi; and Divan, Ryan (November 2026). Radiation Transport Equation. Presented at 2026 UIC not yet.*

Ojha, Sanandan*; Kulkarni, Pranav; Patel, Sharad; Lizdas, Dave; Mesber, Simon; Lampotang, Samsun; and Dooley, Cole (November 2026). Efficacy of a Mixed-Reality Simulator for Ultrasound-Guided Infant Caudal Epidural Block Training: Study Design. Poster presented at 2026 Fall Undergraduate Research Symposium, Gainesville, Fl.*

***Ojha, Sanandan**; Kulkarni, Pranav; Patel, Sharad; Lizdas, Dave; Mesber, Simon; Lampotang, Samsun; and Dooley, Cole (October 2026). Efficacy of a Mixed-Reality Simulator for Ultrasound-Guided Infant Caudal Epidural Block Training: Study Design. Poster presented at Biomedical Engineering Society Annual 2026 Meeting, Orlando, Fl.*

***Ojha, Sanandan**; Nguyen, Thien-Nhi; Shreekanth, Shreya; Shah, Rushi (October 2026). Evaluation of a Kidney Transplant Model for Educational Skill Development. Poster presented at Biomedical Engineering Society Annual 2026 Meeting, Orlando, Fl.*

***Ojha, Sanandan** & Stepien, Tracy (April 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Poster presented at 2026 Spring Undergraduate Research Symposium, Gainesville, FL.*

***Ojha, Sanandan** & Stepien, Tracy (March 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Poster presented at 2026 Infectious Diseases and Economic Growth Workshop, Gainesville, FL.*

***Ojha, Sanandan**; Gill, Jali; and Zhu, Luoding (July 2025). Modeling Cell Migration with a Linearized Lattice-Boltzmann Immersed-Boundary Method. Poster presented at 2025 Indiana University Indianapolis Summer Research Symposium, Indianapolis, IN.*

***Ojha, Sanandan**; Kennedy, Hannah; Sharma, Utsav; and Labasbas, Joshua (April 2025). Synthetic MRI Generation for Deep Brain Stimulation Preoperative Planning. Poster presented at 2025 Yale Undergraduate Research Conference in New Haven, CT.*

***Ojha, Sanandan**; Kennedy, Hannah; Sharma, Utsav; Jain, Ayush; and Kelegama, Ania (November 2023). 3D Modeling of Berlin Heart Devices: Mitigating Pediatric Anxiety in Congenital Heart Disease. Poster presented at 2023 Spring Undergraduate Research Symposium by Center for Undergraduate Research, Gainesville, FL.*

ORAL PRESENTATIONS

***Ojha, Sanandan;** Santiago etc…. DRP presentation …. Late November*

***Ojha, Sanandan;** Dadlani, Rishi; Divan, Ryan; and Ratcliff, Gavin (May 2026). Radiation Transport Equation. Presented at 2026 TAMU Undergraduate Summer School in Modeling and Simulating with PDEs, College Station, TX.*

***Ojha, Sanandan** & Stepien, Tracy (April 2026). Mathematical Modeling of Intercalation in Two Cell Layers During the Gastrulation Stage of Embryonic Development. Presented at 2026 Undergraduate Mathematics Research Symposium, Gainesville, FL.*

***Ojha, Sanandan**; Kulkarni, Pranav; and Patel, Sharad (April 2026). Modeling AI Model Collapse Using Markov Chains and Layered Discrete Distributions. Presented at 2026 Undergraduate Mathematics Research Symposium, Gainesville, FL.*

***Ojha, Sanandan**; Gill, Jali; and Zhu, Luoding (July 2025). Modeling Cell Migration with a Linearized Lattice-Boltzmann Immersed-Boundary Method. Presented at 2025 Indiana Undergraduate Math Conference, Indianapolis, IN.*

***Ojha, Sanandan** & Muthukrishnan, Ashok (October 2022). Utility of SSTR PET Imaging in Management of Olfactory Neuroblastomas. Presented at 2022 Annual Southeastern Chapter of Society of Nuclear Medicine and Molecular Imaging, Savannah, GA.*

PUBLICATIONS

Ojha, Sanandan & Muthukrishnan, Ashok *(July 2022). Pluvicto Approval Launches Theragnostic in Prostate Cancer. Targeted Therapies in Oncology. 11(10):77.*

Manuscripts in Preparation

***Sanandan Ojha**, Jean-Luc Choiseul, Holley E. Lynch, Melis Hazar, Lance A. Davidson, Tracy L. Stepien. Model Selection of Collective Cell Migration in Multi-Layer Aggregates.* In Preparation.

*Luoding Zhu, **Sanandan Ojha,** Jali Gill, Jared Barber. An Implicit Lattice Boltzmann – Immersed Boundary Method for Cell Migration.* In Preparation.

HONORS

University Scholars Program, CLAS, Total $1750 — *Fall ’26—Spring ‘27*
Best Student Presenter, SECSNMMI — *Fall ‘22*
Outstanding SCUDEM X Participant, SIMIODE — *Fall ‘25*
Gold Medal, IGEM — *Fall ‘24*
Silver Medal, IGEM — *Fall ‘25*
Successful Participant MCM/ICM, COMAP — *Spring ‘24, Spring ‘25, Spring ‘26*
Successful Participant SCUDEM, SIMIODE — *Fall ‘24*
UF Math Circle Undergraduate Leader, UF Mathematics — *Spring ‘26*
Undergraduate Research Program, UF Mathematics, $600 — *Spring ‘26*
BME LA Program, College of Engineering, Total $1500 — *Fall ‘24, Fall ‘25*
Dean’s List, CLAS — *(3 awards) Spring ’25—Spring ‘26*
Dean’s List, College of Engineering — *(7 awards) Fall ‘23—Spring ’26*

PROJECTS

CONMED: Senior Design Engineering Final Project — *August 2026—Present*

Distal Bicep Tendon Repair System

- Designing a proof-of-concept distal bicep tendon repair system for our industry sponsor, CONMED, with a team of eight senior engineering students.
- System includes implant, instruments, and sterile packaging, evaluated by a sports medicine surgeon.

Quantitative Physiology: BME4409

Restenosis in Stents — *August 2026—Present*

- Adapting a model for restenosis built for coronary stents to self-expanding nitinol stents in the leg.
- Simplified to a 2D problem, introducing a constant outward pressure inducing stress factors in the femoral artery.

Modeling and Simulating with Partial Differential Equations

Radiation Transport Equation — *May 18—May 29, 2026*

- Developed theory and solver for the Boltzmann Equation of Inertial Confinement Fusion using 1D slab geometry.
- Coupled Streamline Upwind Petrov-Galerkin solver with Richardson Iteration for integro-differential term.
- Solved benchmark problems including the Reed problem and the Grazing problem.

SIMIODE Challenge Using Differential Equations Modeling

Problem B: “An AI Ouroboros” — *October 18—November 11, 2025*

- Constructed a modified Markov Chain and layered discrete distributions to model AI collapse in a task space.
- Made 10-minute video on results, recognized as **Outstanding Participant** and featured on MathWorks website.

Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling

Problem A: “Modeling Smartphone Battery Drain” — *January 29—February 2, 2026*

- Designed **continuous time models** in MATLAB and used **UQ-Lab** for sensitivity analysis.
- Wrote 25-page report on model, recognized as a **Successful Participant**.

Problem A: “The Constant Wear on Stairs” — *January 23—January 27, 2025*

- Created **3 difference equations** to estimate 3 values using limited, measurable parameters.
- Wrote 15-page report on models and was recognized as a **Successful Participant**.

Problem C: “Momentum in Tennis” — *February 1—February 5, 2024*

- Analyzed a data set from the 2024 Wimbledon Championships **to construct** 4 models using **R-Programming**.
- Created a 25-page report describing analysis, findings, and models, recognized as a **Successful Participant**.

Infectious Diseases and Economic Growth: Modeling Early COVID-19 in FL — *March 26—March 29, 2026*

- Created a simplified **SIRS** model with economic interactions in **Python**, fitted against early Florida infection data.
- Used **Approximate Bayesian Computation** to reduce parameter space and compare likely long-term behavior.

PCB3713C Auto-Sorter — *September 2025—October 2025*

- Created an executable to format student submissions for uploading to Canvas using **Python**.

Dream Team Engineering Bias Detection *August 2025—September 2025*

- Analyzed 8 cycles of past application data and acceptances for Dream Team Engineering using **R Programming**.
- Looked for bias and trends in individual reviewer decisions and overall application pool.

Herbert Wertheim College of Engineering Outreach Design Competition *February 17, 2026*

- Designed 3D printed mobile cranes with hydraulic booms for 6th-8th graders using **Fusion 360**.
- Created matching handouts to illustrate concepts relating to Pascal's Law and Mechanical Advantage to students.

Dream Team Engineering Designathon

Multi-Purpose Piston-Powered Walker *November 7—November 9, 2025*

- Designed a modified walker that provides powered assistance to the user in traversing stairs and getting seated.
- A 3D CAD of the walker was created using **Fusion 360**, along with budget and stakeholder analysis.

Prescription Puzzles *October 12—October 13, 2023*

- Created a low-cost pill dispenser using a single **Arduino** that is activated by completion of a Jigsaw puzzle.
- Physical prototypes of both devices were created.

UCSAS USOPC Data Challenge *November 1, 2023—January 15, 2024*

- Chose 10 athletes for US Olympic Gymnastics team by analyzing collated historical event data in Python.
- Wrote a short report detailing methodology and conclusions.

TEACHING EXPERIENCES

Undergraduate Teaching Assistant

CHM2100: Organic Chemistry *May 2026—August 2026*

- Created a discord environment for students to engage TAs and study, with over **100** students participating.
- Held weekly review sessions covering material from the week as well as test practice.

PCB3713C: Cellular and Systems Physiology *August 2025—August 2026*

- Designing new labs and activities to accommodate expanded class times for Fall 2026.
- Ran weekly student labs, held office hours, provided mentorship to student groups, and graded assignments.

BME Learning Assistant

BME3053C: Computer Applications for BME *February 2026—May 2026*

- Hosted office hours, helped students with homework, and provided mentorship as a peer mentor.

BME3060: Biomedical Engineering Fundamentals *February 2025—May 2025*

- Hosted office hours, created exam reviews, and provided mentorship as a peer mentor.

LEADERSHIP

MATLAB Club *August 2026—Present*

- Hold events promoting the use of MATLAB for students, including an info session for the SCUDEM competition.

Dream Team Engineering

Director of Research *May 2025—May 2026*

- Directly oversaw and managed **5** undergraduate research teams with **34** members for the academic year.
- Evolved 'Research Bootcamp' to showcase fundamentals of ANOVA analysis and Machine Learning principles.

Assistant Director of Research *August 2024—May 2025*

- Directly oversaw and managed **4** undergraduate research teams with **29** members for the academic year.
- Ideated and held 'Research Bootcamp' to introduce fundamentals of research and scientific writing.

Badminton Club

Executive Board Member *August 2025—Present*

- Organized volunteer events for the UF Badminton Club, creating community and helping the larger Alachua area.

- Safety Officer in charge of first response of health emergencies at club practices, tournaments, and other events.

BME Inclusive Excellence

Secretary *August 2024—May 2025*

- Increased undergraduate participation in BME inclusive excellence events, fostering greater community in BME.
- Designed an event for World Mental Health Day, highlighting mental health among academics and students.

STEM OUTREACH

University of Florida, Math Circle *August 2025—Present*

- Helping elementary - high school students learn problem-solving skills by tackling math games and problems.
- Designed and hosted Math Circle meeting about teaching Monte Carlo estimation and probability.

University of Florida, MentorGNV *August 2025—May 2026*

- Helping elementary/middle school students in after and during school activities, including tutoring in prealgebra.

Event Volunteering

University of Florida, Boo at the Zoo	*October 28, 2027*
University of Florida, Math Festival	*March 29, 2026*
Alachua, STEAM Night	*March 27, 2026*
Indiana University Indianapolis, STEM YES Engineering/Technology Event	*June 18, 2025*
Alachua, Regional STEM Fair	*January 5, 2025*
University of Florida, Florida Science Olympiad State Event	*April 6, 2024*

COMMUNITY SERVICE

University of Florida, Badminton Club

Jimmy Carter Invitational	*2025—2026*
NAIA Track and Field	*2025—2026*
Keep Alachua County Beautiful	*2024—2026*
Food4Kids	*2026*
Gator Slam	*2023—2025*
Gator Smash	*2024—2026*

CLINICAL EXPERIENCE

West Palm Beach, FL, Florida Theragnostic Center *May 2026—August 2026*

Supervisor: Dr. Krishnan

- Shadowed nurses, technicians, and radiologists throughout workday.

West Palm Beach, FL, VA Medical Center *June 2022—August 2022*

Supervisor: Dr. Mullin

- Shadowed multiple radiologists throughout workday.

Supervisor: Dr. Shetty

- Shadowed oncologist, dosimetrist, and physicist throughout workday.

SKILLS

Languages: Python, MATLAB, R-Programming, React, Java, LaTeX/Markdown.

Technical Skills: NetLogo, Arduino IDE, KiCad, Fusion 360, ImageJ, UQ-Lab, BindCraft, AlphaFold, HiPerGator.

Wet-Lab: Cell Culturing, Protein Purification, Bacterial Transformation, Overexpression, SDS-PAGE.

Modeling: Continuum Mechanics, Lattice Boltzmann–Immersed Boundary, Agent-Based Modeling, Approximate Bayesian Computation.

Certification: American Red Cross Basic Life Support